

TEACHING CASE

Collision tumor of malignant skin cancers: A case of melanoma in basal cell carcinomaGiovanni Papa^a, Giorgio Grandi^b, Michele Pascone^{a,*}^a*Department of Plastic and Reconstructive Surgery, University of Trieste, Ospedale di Cattinara, strada di Fiume 447, 34100 Trieste, Italy*^b*Department of Pathological Anatomy, University of Trieste, Ospedale Maggiore, piazza Ospedale 1, 34100 Trieste, Italy*

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Abstract

The coexistence of two malignant skin tumors intermingling in the same histologic specimen is rare. We report a case of melanoma in a basal cell carcinoma collision tumor. The presence of two intermingled neoplasms was confirmed by immunohistochemistry, which showed strong positivity for S-100 and HMB-45. A review of the literature and a case analysis allow for some pathogenetic and prognostic considerations.

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Introduction

Although rare, the association of two different skin tumors in the same histologic specimen has been described by many authors [14]. In most cases, the lesions are found to be contiguous, up to 1 mm apart [3]. Far less frequent is the collision of two tumors. Collision refers to the coexistence of two tumors intermingling within the same lesion [13].

The collision of malignancies originating from different cell lines in a single skin lesion is a rare occurrence.

In particular, the collision of basal cell carcinoma and malignant melanoma (MM) is very rare, although both lesions may develop in skin damaged by chronic exposure to sunlight [11].

The pathogenesis of coexisting skin tumors remains a matter of speculation [5].

Some authors consider the various combinations to be merely coincidental [2,12], whereas others hypothesize that the presence of one tumor may induce epithelial or stromal changes responsible for the development of a second tumor [4,15,19].

Basal cell carcinomas (BCC) are considered tumors of trichogenic origin in most cases, i.e., they derive from the hair follicle of the skin [10]. Hair follicles as well as BCC are known to contain HMB45-positive melanocytes [9], which may give rise to clinically pigmented lesions [8].

Therefore, an abnormal stimulation of such melanocytes could lead to the development of MM within BCC [7].

This is most possibly the reason for the rarity of a “basal cell carcinoma-associated MM”.

A review of the literature revealed a dearth of reports on MM arising in basal cell carcinoma (Table 1).

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Table 1. Literature describing cases with collision between a basal cell carcinoma (BCC) and a malignant melanoma (MM)

Author/s	Observation	Reference
Kao (1983)	1 case of MM in BCC	Annual meeting of the American Society of Dermatopathology [2]
Bhawan et al. (1984)	2 cases, 68 and 56 years old; 2×1 cm and 1.5×1.2 cm, superficial MMIS in BCC of the back	[6]
Coskey and Mehregan (1987)	1 case, 50 years old; non-melanotic nodular MM in a BCC of the back, 0.7 mm thick	[18]
Sina and Samarodin (1989)	1 case, 83 years old; nodular BCC in malignant lentigo	[5]
Burkhalter and White (1997)	1 case, 69 years old; superficial MMIS in BCC of the ear, 1.7 mm thick	[12]
Hirakawa et al. (1998)	1 case, 78 years old; MM in BCC of the temporal region, pT4a with Breslow 9 mm confirmed by PCR	[20]
Wang et al. (2003)	1 case, 72 years old; malignant lentigo in a recurrence of BCC on the eyelid	

The earliest case reported in the literature dates back to 1983 and was presented by Kao at the Annual Meeting of the American Society of Dermatopathology. Two cases of collision of superficial melanoma in basal cell carcinoma of the back were described in 1984 [2].

In 1987, Coskey reported a case of non-melanotic nodular melanoma in a basal cell carcinoma of the back [6], and in 1989, Sina described a case of basal cell carcinoma in a malignant lentigo of the face [18].

More recent cases include a superficial melanoma in a basal cell carcinoma of the ear, reported in 1997 [5], and a collision between melanoma and basal cell carcinoma in the temporal region, reported in 1998, in which the different nature of the two lesions was confirmed by PCR [12].

The most recent case report, published in 2003, describes a malignant lentigo developing in a recurrence of basal cell carcinoma [20]. In a retrospective review of 78,000 excision biopsies, published in 1997, Piérard identified 11 cases of contiguous cutaneous lesions which, being separate from one another as shown in the figures, cannot be defined as true collision tumors [14].

Case report

A 68-year-old woman presented with a 5-cm polypoid pigmented cutaneous neof ormation in the laterocervical region, which had been present for several years.

The patient also had two other neof ormations: one, also a polypoid pigmented lesion, was located on the back and was histologically diagnosed as a primitive melanoma (epithelioid cell melanoma with a maximum vertical thickness of 2.5 cm and Clark 5), and the other was a squamous cell carcinoma on the cheek. The precise chronology of the appearance of the lesions could not be determined owing to the patient's poor reliability. However, the lesions must have been present

for several years before the patient eventually decided to seek medical attention.

The clinical diagnosis of the laterocervical lesion proved difficult because of its irregular margins, ulceration, and the irritation caused by the position of the exophytic portion at the tumor base.

The lesion was excised and the area repaired by direct skin closure.

Histology revealed the presence of two different tumors in the same lesion: a basal cell carcinoma and a pigmented epithelioid melanoma combined in the same specimen (Fig. 1).

The first routine histologic test with hematoxylin and eosin had already demonstrated the presence of a second tumor inside the basal cell carcinoma, giving rise to the exophytic portion of the lesion (Fig. 2).

The suspicion that the second tumor was a melanoma with a maximum vertical thickness of more than 2.5 cm and Clark 5 (T_4b N_x M_x) was confirmed by immunohistochemistry, which showed strong positivity for S-100 and HMB-45 and therefore revealed the presence of two intermingling neoplasms (Fig. 3).

Given that the patient came to our attention many years after their development, we were not able to determine which of the two lesions developed first, the time that has elapsed between the formation of the two, or if they were contiguous or overlapped.

The patient was treated with interferon due to the advanced stage of the disease.

Discussion

This report describes a further case of collision between basal cell carcinoma and melanoma. Even in our case was the lesion not located on the face, but in the laterocervical region. Most probably, its location in a poorly visible area allowed our patient to hide the disease from her family and doctor.

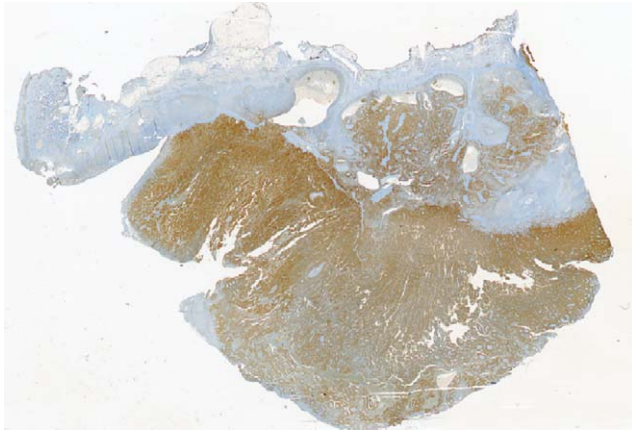


Fig. 1. Original magnification ($\times 1$); immunocytochemical staining (HMB-45) shows a basal cell carcinoma at the base of the lesion and a pigmented epithelioid melanoma on its exophytic portion.

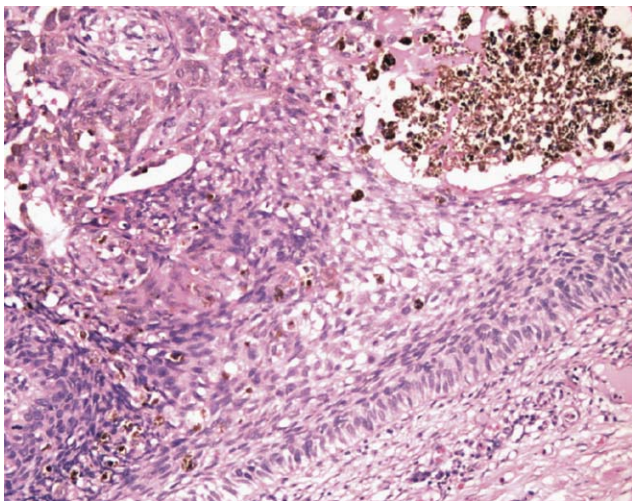


Fig. 2. High magnification with hematoxylin and eosin ($\times 200$) reveals a combination and a focal connection of a typical basal cell carcinoma with malignant melanocytes of melanoma (top left) surrounding it.

Given the small number of cases reported in the literature, it is still difficult to predict the biologic evolution of these lesions. In pathogenetic terms, the long evolution of the tumor could lead to the assumption that the melanoma developed from the nests of melanocytic cells within the basal cell carcinoma by chance [12] or as a result of neoplastic stimuli. On the other hand, the existence of more than one tumor in our patient suggests reduced immune activity and therefore a greater susceptibility to the development of tumors [11,16].

Taking into account the origin of the basalioma and the presence of melanocytes, we could add the rare MM variants, such as “BCC-associated MM” (apart from a mucous membrane MM, etc.), to the already well-

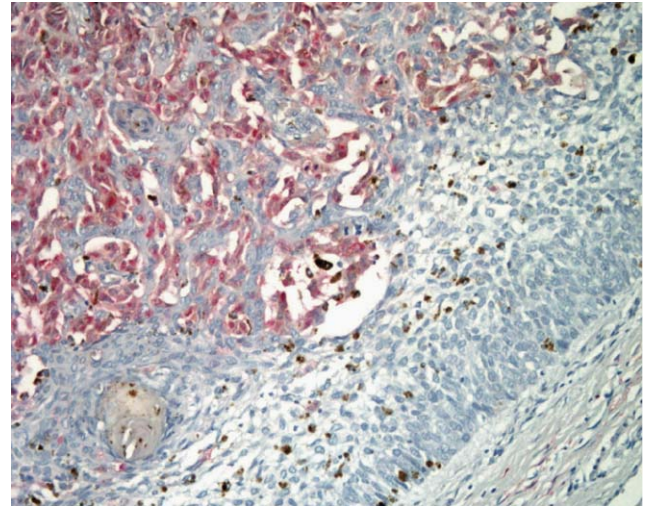


Fig. 3. The immunohistochemical marker S-100 protein ($\times 200$) stained the atypical melanocytes in red and showed the intermingled neoplasms more intensely.

known subtypes of MM (i.e., superficial spreading melanoma, nodular MM, acro-lentiginous MM, nevus-associated MM).

In agreement with other authors, we believe that melanomas arising within less aggressive lesions should also be considered less aggressive, because the tumor in which they have arisen may act as a barrier [5]. This view is supported by the long time elapsed between the development of the tumor and excision of the lesion in our patient. Unfortunately, the presence of a second, probably also primary, melanoma and the impossibility of establishing the chronology of the lesions do not allow for an evaluation of the prognosis of the melanoma colliding with the basal cell carcinoma, although the patient reported that it had grown over several years.

In our opinion, the BCC melanoma cannot be considered metastasis, because the second melanoma (of the back), though being at risk of developing into metastasis, did not give rise to other metastases (for instance cutaneous ones). The second reason is the rarity of a malignant tumor metastasizing into a benign one. Metastasis of a malignant tumor into another malignant tumor is an exceptionally rare event [1].

The best treatment strategy is also difficult to define because of the complexity of the diagnostic approach to this type of lesions.

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